E-COMMERCE DATASET PROBLEM STATEMENTS

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PROBLEM STATEMENT 1

CUSTOMER LOYALTY & REPEAT PURCHASES

Problem Statement :-

What is the relationship between **Customer_Loyalty_Points** and **repeat purchases (Previous_Orders)?** Do customers with higher loyalty points tend to have a higher order frequency?

Problem Statement 1 — Steps:-

- Select required columns
- Aggregate
- Correlation
- Segmentation → Low | Medium | High | Very High
- Comparison
- Visualization
- Insight

PROBLEM STATEMENT 2

CUSTOMER SATISFACTION

Problem Statement: -

Which factors — **Shipping Days, Discount %, Product Category,** and **Delivery Rating** — most strongly influence **Customer Satisfaction?**

Problem Statement 2 — Steps:-

- Select Variables required from data frame
- Correlation Analysis
- Category Analysis
- Identify Strongest Factors
- Visualization
- Business Insight

PROBLEM STATEMENT 3

MARKETING SOURCE & CUSTOMER VALUE

Problem Statement:-

Which **Marketing_Source** brings in the most valuable customers based on **order value, loyalty points,** and **repeat purchases**?

Problem Statement 3 — Steps:-

- Select Variables
- Group by Marketing Source
- Calculate Metrics
- Compare Sources
- Visualization
- Business Insight

PROBLEM STATEMENT 4

DISCOUNT IMPACT ON BUSINESS PERFORMANCE

Problem Statement :-

How do **discounts** affect **Net Sales, Profit,** and **Profit Margin?**

Steps :-

1. Select Variables
2. Create Discount Groups
3. Calculate Metrics
4. Correlation Analysis → Discount % ↔ Sales, Profit & Profit Margin
5. Compare Discount Groups
6. Visualization
7. Business Insight

PROBLEM STATEMENT 5

PRODUCT REVENUE & PROFITABILITY

Problem Statement :-

Which **products** generate **high revenue** but **low profit**, and which products have the best **profit margins**?

Steps :-

1. Select Variables
2. Group by Product
3. Calculate Metrics → Total Revenue, Total Profit, Average Profit Margin, Total Quantity
4. Rank Products
5. Classify Products
6. Visualization
7. Business Insight

THANK YOU